

RESIDUAL SPECTRAL GRAPH LEARNING FOR CONNECTIVITY-LOSS ESTIMATION UNDER MULTI-EDGE DISRUPTIONS

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ABSTRACT. We estimate algebraic-connectivity loss after simultaneous road-network edge disruptions. The proposed model learns a bounded graph-neural correction to an interpretable first-order Fiedler approximation. We compare eight learned configurations and the analytical baseline under independent and spatially clustered failures, using graph-disjoint synthetic splits and zero-shot transfer to five OpenStreetMap (OSM) networks. Results are replicated with five seeds and uncertainty is computed by resampling seeds, not individual scenarios. On OSM, the residual model obtains mean absolute error (MAE) 0.097 (95% seed-bootstrap CI 0.091–0.102) for independent failures and 0.090 (0.071–0.124) for spatial failures. Relative to a direct GCN, paired MAE improvements are 0.0085 (0.0039–0.0131) and 0.0121 (0.0026–0.0247), respectively. Fiedler-node and coordinate ablations are inconclusive, indicating that the explicit residual prior provides the robust gain. Exact OSM eigendecomposition averages 5.09 ms per scenario; the first-order update requires 0.018 ms and learned inference 0.44–0.67 ms on a CPU. This is structural disruption analysis, not hazard or traffic-flow prediction.

1. INTRODUCTION

Removing a few links can change global road-network connectivity. Exact spectral recomputation is reliable but may be expensive in large scenario studies, while a local sensitivity is fast but ignores finite interactions. Graph neural networks (GNNs) can represent such interactions, yet direct end-to-end learning may discard useful mathematics. We therefore ask whether a GNN should predict connectivity loss directly or correct a spectral prior.

Our contributions are: (i) a residual spectral-GCN for multi-edge disruption; (ii) separation of an explicit spectral prior from Fiedler node features; (iii) independent and spatial-cluster failures on five OSM networks; (iv) direct GCN, Deep Sets, summary-MLP, feature, and residual ablations; and (v) seed-level paired uncertainty, runtime, and regime-specific error analysis.

2. RELATED WORK AND NOVELTY POSITIONING

Fiedler’s algebraic connectivity λ_2 characterizes connectedness and underlies spectral robustness analysis [1, 2]. Its eigenvector gives an edge-weight derivative, but summing derivatives is a local approximation and need not capture simultaneous finite deletions.

Road robustness has been studied under random, targeted, and localized link failures. Synthetic road graphs permit controlled topology experiments [3]; hazard-independent multi-link analysis also demonstrates that conclusions depend on disruption multiplicity [4]. Our target is structural connectivity, complementary to travel time, flow, or accessibility and not a substitute for them.

GCNs propagate local information over graph adjacency [5], whereas Deep Sets supplies a permutation-invariant baseline without message passing [6]. Physics-informed graph learning combines analytical structure and learned components [7]. We do not claim the first use of GNNs or spectral robustness. The novelty is the controlled, leakage-resistant comparison of node-level spectral information versus an explicit residual prior, across two failure processes and five unseen OSM networks, with paired seed-level inference and computational profiling.

3. FORMULATION AND MODELS

Let $G = (V, E, w)$ be connected and undirected, $L = D - A$ its combinatorial Laplacian, and $0 = \lambda_1 \leq \lambda_2 \leq \dots$ its eigenvalues.

Definition 3.1 (Relative connectivity loss). *For disrupted edges $S \subseteq E$,*

$$(1) \quad y(G, S) = \text{clip}(1 - \lambda_2(G - S)/\lambda_2(G), 0, 1).$$

Disconnection therefore gives $y = 1$.

If u_2 is a unit Fiedler vector and λ_2 is simple, then $\partial\lambda_2/\partial w_{ij} = (u_{2,i} - u_{2,j})^2$. Our analytical baseline is

$$(2) \quad \hat{y}_{\text{spec}} = \text{clip} \left(\frac{\sum_{(i,j) \in S} w_{ij} (u_{2,i} - u_{2,j})^2}{\lambda_2(G)}, 0, 1 \right).$$

The ScenarioGCN receives the damaged adjacency and node features comprising normalized original degree, disrupted-edge incidence, absolute normalized Fiedler coordinate, and two normalized spatial coordinates. Three 48-unit GCN layers are followed by mean-max pooling and a two-layer head. The residual variant predicts

$$(3) \quad \hat{y}_{\text{res}} = \text{clip}(\hat{y}_{\text{spec}} + \tanh r_{\theta}(G, S), 0, 1).$$

We compare direct and residual versions with all features, without Fiedler, and without coordinates. Additional baselines are a Deep Sets node encoder and an MLP using graph summary statistics. All learned models use the same split, AdamW, mean-squared error, at most 80 epochs, and validation early stopping.

4. EXPERIMENTAL DESIGN

4.1. Synthetic data and failure processes. Connected random geometric graphs contain 35–65 nodes with inverse-length edge conductance. Each seed has 1,600 balanced scenarios (800 per failure mode): 1,120 train, 240 validation, and 240 test examples, split by base-graph identifier. No damaged version of a training graph can enter validation or test. Independent failure samples 1–8 edges uniformly. Spatial failure chooses a random epicentre and removes the requested number of edges whose midpoints are nearest to it.

4.2. Five-network OSM transfer. OSMnx [8] downloaded 650-m drivable networks around five Vietnamese sites. Directed multigraphs were simplified, converted to undirected simple graphs, and restricted to their largest connected component. Conductance is $100/\text{length}_m$. Table 1 spans different sizes and densities. For every seed, site, and failure process, 100 scenarios were generated, giving 1,000 OSM scenarios per seed. OSM data are never used for training, early stopping, or model choice.

TABLE 1. Cached OSM networks. Density is $2|E|/(|V|(|V|-1))$.

Site	Nodes	Edges	Density
HCMUS (Ho Chi Minh City)	163	221	0.0167
VIASM (Hanoi)	206	301	0.0143
Da Nang	142	203	0.0203
Can Tho	114	178	0.0276
Da Lat	48	55	0.0488

4.3. Statistics and runtime. Metrics are calculated within each seed. Reported 95% intervals resample the five seed-level values 10,000 times. Comparisons use within-seed paired differences; scenarios sharing a graph or seed are never treated as independent replicates. Runtime uses one PyTorch CPU thread on the same 16-logical-CPU Windows machine, and includes per-scenario prediction but excludes training and data loading.

5. RESULTS

TABLE 2. OSM zero-shot results: seed mean MAE [95% seed-bootstrap CI] and Spearman correlation.

Method	Independent		Spatial cluster	
	MAE	ρ	MAE	ρ
Spectral first order	0.502 [0.482,0.521]	0.391	0.650 [0.643,0.657]	0.324
Summary MLP	0.281 [0.240,0.320]	0.819	0.347 [0.269,0.425]	0.787
Deep Sets	0.114 [0.105,0.123]	0.765	0.143 [0.122,0.164]	0.789
Direct GCN	0.106 [0.100,0.112]	0.848	0.102 [0.077,0.148]	0.841
Residual GCN	0.097 [0.091,0.102]	0.903	0.090 [0.071,0.124]	0.896
Residual, no Fiedler	0.096 [0.090,0.101]	0.901	0.078 [0.067,0.090]	0.889
Residual, no coordinates	0.098 [0.092,0.102]	0.901	0.095 [0.073,0.122]	0.895

Residual-full improves upon direct-full by paired MAE 0.0085 [0.0039,0.0131] for independent and 0.0121 [0.0026,0.0247] for spatial failures. On synthetic graphs the improvements are 0.0253 [0.0106,0.0400] and 0.0160 [0.0048,0.0241]. Thus the residual formulation is the stable contribution. In contrast, every paired MAE CI comparing residual-full with its no-Fiedler or no-coordinate variant contains zero. The spatial OSM point estimate even favours no-Fiedler. These features should be regarded as optional under the present power, not established necessities.

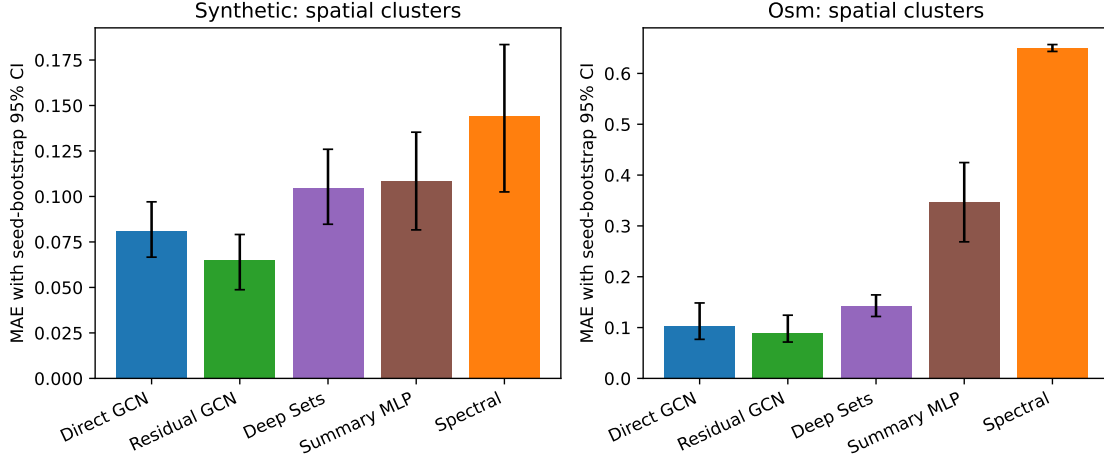


FIGURE 1. Spatial-cluster performance in the synthetic and OSM domains. Error bars are 95% intervals from seed-level bootstrap.

TABLE 3. Mean per-scenario runtime on OSM (milliseconds).

Method	ms	Method	ms
Exact eigendecomposition	5.094	Spectral first order	0.018
Summary MLP	0.442	Deep Sets	0.430
Direct GCN	0.502	Residual GCN	0.674

5.1. Runtime. The analytical approximation is about 287 times faster than exact recomputation; residual inference is about 7.6 times faster. These ratios are implementation- and graph-size-specific. Mean training time ranges from 60 s (summary MLP) to 150 s (slowest GCN ablation), and residual-full averages 121 s per seed on CPU.

5.2. Error regimes. For residual-full on pooled OSM failure modes, seed-mean MAE is 0.070 for connected damaged graphs and 0.112 after disconnection. By failed-edge count it is 0.090 (one), 0.116 (two–three), 0.089 (four–five), and 0.081 (six or more), so error is not monotone in disruption count. Site MAE is 0.041 Can Tho, 0.061 VIASM, 0.088 Da Nang, 0.089 HCMUS, and 0.189 Da Lat. The dense but much smaller Da Lat graph is hardest, warning against attributing error to density alone. No evaluated scenario clipped Equation (2) at one; consequently the requested clipped-versus-unclipped comparison is undefined rather than silently omitted. The poor spectral OSM MAE instead reflects large non-saturating overestimates.

6. DISCUSSION AND LIMITATIONS

The explicit prior substantially improves a direct GCN even when the prior alone transfers poorly. It supplies useful rank structure, while the learned correction calibrates nonlinear finite-deletion effects. Deep Sets is competitive only in selected regimes and the summary MLP loses substantial accuracy, supporting the use of graph structure.

Five seeds permit uncertainty reporting but not high-powered claims about small feature effects. The OSM samples cover only 650-m neighbourhoods in Vietnam; Da Lat also differs in size as well as density. Spatial clusters are geometric proxies, not measured floods, landslides, or construction events. Algebraic connectivity

ignores traffic demand, direction, capacity, travel time, and recovery. Future work should use larger networks, blocked geographic transfer, licensed hazard layers, and sparse/iterative eigensolvers.

7. CONCLUSION

Across both disruption processes and both domains, learning a correction to a Fiedler approximation is more reliable than direct GCN regression. Full feature ablation does not support strong claims for Fiedler node coordinates or spatial coordinates individually. Multi-area transfer, seed-paired statistics, runtime profiling, and explicit failure-regime analysis make the result suitable as a reproducible poster study while delimiting its real-world interpretation.

REPRODUCIBILITY STATEMENT

The repository includes cached GraphML, all generators and models, raw-output schemas, a 10,000-resample analysis script, and `reproduce_all.ps1`. Seeds are 11, 22, 33, 44, 55. The recorded environment is Python 3.13.11; NumPy 2.4.4; SciPy 1.17.1; NetworkX 3.6.1; pandas 2.2.3; scikit-learn 1.9.0; PyTorch 2.11.0 (CPU); Matplotlib 3.11.1; OSMnx 2.1.1; and GeoPandas 1.1.4. Hardware is an AMD64 Family 25 Model 68 system with 16 logical CPUs; CUDA was unavailable and no GPU was used. OSM data are copyright OpenStreetMap contributors under the ODbL [9].

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